ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Light - Reflection and Refraction** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** second law of reflection	Statement of the second law.
2	**(c)** in vacuum to that in the medium	n = c/v.
3	**(c)** a converging lens	It converges parallel rays.
4	**(c)** virtual, erect and of the same size as the object	Plane mirror image is virtual, erect, same size, laterally inverted.
5	**(a)** bends towards the normal	Glass is optically denser.
6	**(a)** +2 D	P = 1/0.5 m = 2 D.
7	**(c)** pole	The pole lies on the mirror.
8	**(a)** bends away from the normal	Speed increases.
9	**(b)** dioptre	1 D = 1 m^-1.
10	**(a)** R/2	f = R/2.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Angle of incidence equals the angle of reflection (1); Incident ray, reflected ray and normal lie in the same plane (1).	1+1
12	Concave mirror: negative (1); Convex mirror: positive (1).	1+1
13	The perpendicular distance between the original ray and the emergent ray (1); Seen in a rectangular glass slab (1).	1+1
14	Rays parallel to the principal axis meet at a point after reflection (1); This point on the axis is the principal focus (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Virtual and erect (1); Same size as the object (1); Laterally inverted; as far behind as the object is in front (1).	1+1+1
16	1/v = 1/f + 1/u = 1/10 - 1/30 (1); v = +15 cm (1); m = v/u = -0.5 (1).	1+1+1
17	Speed decreases (1); Wavelength decreases (1); Frequency remains the same (1).	1+1+1
	OR: 1/v = 1/f - 1/u = -1/15 + 1/30 = -1/30 (1); v = -30 cm (in front of the mirror) (1); Real, inverted and same size (m = -1) (1).	1+1+1
18	1/v = 1/f + 1/u = -1/20 - 1/20 (1); 1/v = -1/10 (1); v = -10 cm (virtual image on the same side) (1).	1+1+1
	OR: Negative m: real and inverted (1); Positive m: virtual and erect (1); Magnitude greater than 1 means enlarged (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	At infinity: at F, real, point-sized (1); Beyond C: between F and C, real, inverted, diminished (1); At C: at C, real, inverted, same size (1); Between C and F: beyond C, real, inverted, enlarged; at F: at infinity, enlarged (1); Between F and P: behind the mirror, virtual, erect, enlarged (1).	1+1+1+1+1
	OR: Power = 1/f in metres; unit dioptre (1); P1 = +4 D (1); P2 = -2 D (1); Net power = +2 D (1); Combination is converging with f = 50 cm (1).	1+1+1+1+1
20	Mirror formula 1/v + 1/u = 1/f (1); 1/v = -1/20 + 1/25 = -1/100 (1); v = -100 cm (1); m = -v/u = -4, so h' = -16 cm (1); Image real, inverted, magnified (1).	1+1+1+1+1
	OR: Concave in shaving, dentist, headlights (1); Convex in rear-view mirrors (1); Bulb at the focus of a concave mirror gives a parallel beam (1); Convex mirror spreads light over a wide area (1); Convex mirror gives a wider field of view (1).	1+1+1+1+1

Mistakes to avoid:

- Ignoring the sign convention: for a concave mirror or lens use $u < 0$ and, for a concave mirror, $f < 0$ (convex lens $f > 0$, concave lens $f < 0$).
- Using the lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ for mirrors, or the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ for lenses.

- Using f in centimetres while calculating power; $P = 1/f$ gives dioptres only when f is in **metres**.
- Saying that light bends away from the normal when entering glass; it bends towards the normal in a denser medium.
- Describing the image formed by a concave mirror when the object is between P and F as real; it is virtual, erect and enlarged.
- Writing the refractive index as $\sin r / \sin i$; Snell's law gives $n = \sin i / \sin r$ for light going from air into the medium.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Light - Reflection and Refraction** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 2×10^8 m/s	v = c/n.
2	**(a)** negative	Concave lens has negative focal length.
3	**(a)** at the centre of curvature	At C, the image is at C and of the same size.
4	**(d)** constant	This constant is the refractive index.
5	**(b)** at 2F	Image is also at 2F on the other side.
6	**(c)** between the pole and the focus	Used as a shaving mirror.
7	**(b)** lateral displacement of the emergent ray	The emergent ray is parallel but shifted.
8	**(b)** between optical centre and F	A virtual, erect, magnified image forms.
9	**(a)** Concave mirror	Teeth lie between pole and focus.
10	**(d)** Diamond	Diamond has n = 2.42.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Power of a lens of focal length 1 metre (2).	2
12	Virtual and erect (1); Diminished (1).	1+1
13	Concave mirror in headlights (1); Convex mirror in rear-view mirrors (1).	1+1
14	sin i / sin r is a constant for a given pair of media (1); For light of a given colour (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	f = R/2 = -20 cm (1); Rays from infinity are parallel (1); Image forms at the focus, 20 cm in front of the mirror (1).	1+1+1
16	Power = reciprocal of focal length in metres (1); P = 1/(-0.5) (1); P = -2 D (1).	1+1+1
17	Real image can be obtained on a screen; virtual cannot (1); Real image: concave mirror with object beyond F (1); Virtual image: plane mirror (1).	1+1+1
	OR: Rays diverge after reflection (1); Reflected rays appear to meet behind the mirror (1); Image is virtual, erect and magnified (1).	1+1+1
18	Shaving or make-up mirror (1); Headlights of vehicles and torches (1); Dentist's mirror and solar furnaces (1).	1+1+1
	OR: Rays from the coin bend away from the normal on going into air (1); The rays appear to come from a higher point (1); So the coin looks raised (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Bending of light as it passes from one medium to another (1); Incident ray, refracted ray and normal lie in one plane (1); sin i / sin r = constant (Snell's law) (1); n = c/v = 3×10^8 / 2×10^8 (1); n = 1.5 (1).	1+1+1+1+1
	OR: 1/v = -1/15 + 1/10 = 1/30 (1); v = +30 cm (1); m = -v/u = -30/-10 = +3 (1); Virtual, erect (1); Magnified three times (1).	1+1+1+1+1
20	Ray bends towards the normal at the first surface (1); It travels in a straight line in the glass (1); It bends away from the normal at the second surface (1); Opposite faces are parallel, so the angle of emergence equals that of incidence (1); Hence the emergent ray is parallel but laterally shifted (1).	1+1+1+1+1
	OR: n = sin 60 / sin 30 (1); n = ($\sqrt{3}$/2)/(1/2) = $\sqrt{3}$ (1.73) (1); v = c/n (1); v = 3×10^8 / 1.73 (1); v = 1.73×10^8 m/s (1).	1+1+1+1+1

Mistakes to avoid:

- Ignoring the sign convention: for a concave mirror or lens use $u < 0$ and, for a concave mirror, $f < 0$ (convex lens $f > 0$, concave lens $f < 0$).
- Using the lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ for mirrors, or the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ for lenses.
- Using f in centimetres while calculating power; $P = 1/f$ gives dioptres only when f is in **metres**.

- Saying that light bends away from the normal when entering glass; it bends towards the normal in a denser medium.
- Describing the image formed by a concave mirror when the object is between P and F as real; it is virtual, erect and enlarged.
- Writing the refractive index as $\sin r / \sin i$; Snell's law gives $n = \sin i / \sin r$ for light going from air into the medium.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Light - Reflection and Refraction** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** a virtual, erect and diminished image	It does so for every object position.
2	**(a)** give a wider field of view and erect, diminished images	Convex mirrors always give virtual, erect and diminished images.
3	**(b)** 3/4	It is the reciprocal.
4	**(d)** real and inverted	Negative m means a real, inverted image.
5	**(a)** retraces its path	It strikes the mirror normally.
6	**(c)** refraction of light	Rays bend at the surface.
7	**(b)** 1/v - 1/u = 1/f	Standard lens formula.
8	**(c)** the focus	By definition of the principal focus.
9	**(d)** the nature of the medium	It is a property of the medium.
10	**(a)** goes undeviated	Thin lens property.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	1/v - 1/u = 1/f (1); m = v/u (1).	1+1
12	It always gives an erect image (1); Its wide field of view shows more traffic (1).	1+1
13	Ratio of speed of light in vacuum to that in the medium (1); n = c/v (1).	1+1
14	It converges parallel rays to a point (1); The point is the principal focus (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	n = c/v (1); v = 3×10^8 / 1.5 (1); v = 2×10^8 m/s (1).	1+1+1
16	Rear-view mirror in vehicles (1); Street light reflectors (1); Security mirrors in shops (1).	1+1+1
17	Powers add: P = +5 D (1); f = 1/P = 0.2 m (1); f = 20 cm (1).	1+1+1
	OR: Water is denser, so it bends towards the normal (1); Refracted angle is less than 45 degrees (1); n = sin i / sin r (1).	1+1+1
18	Ray bends towards the normal on entering (1); It bends away on leaving (1); Emergent ray is parallel to the incident ray but displaced sideways (1).	1+1+1
	OR: Object at 2F, i.e. 30 cm (1); Image also forms at 2F (1); Image at 30 cm on the other side (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	At infinity: at F2, real, point (1); Beyond 2F1: between F2 and 2F2, real, inverted, diminished (1); At 2F1: at 2F2, real, inverted, same size (1); Between F1 and 2F1: beyond 2F2, real, inverted, enlarged; at F1: at infinity (1); Between F1 and O: same side, virtual, erect, enlarged (1).	1+1+1+1+1
	OR: A parallel ray diverges away from the principal focus (1); A ray through the optical centre goes straight (1); Image is virtual, erect and diminished (1); Image forms between O and F (1); Uses: spectacles for myopia and in peepholes (1).	1+1+1+1+1
20	1/v = 1/20 - 1/30 = 1/60 (1); v = +60 cm (1); m = v/u = 60/(-30) = -2 (1); h' = -10 cm (1); Real, inverted and magnified (1).	1+1+1+1+1
	OR: All distances are measured from the pole or optical centre (1); Distances along the incident direction are positive (1); Distances opposite are negative (1); Heights above the axis are positive, below negative (1); Focal length is negative for a concave mirror and a concave lens (1).	1+1+1+1+1

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